

9M337 Sosna-R

The **9M337 Sosna-R** (Pine) (SA-X-25)^{[1][2]} is a Russian radar and laser-guided supersonic (Mach 2.6) two-stage missile. It is used in Sosna-R short range air defense missile system designed to protect military units from air attacks in all types of combat situations, including during march.

In 2017, official tests of the newest air defense missile system Bagulnik (domestic variant which is currently named Strela-10ML) were successfully completed.^{[3][4][5]}

Description

It was developed by KB Tochmash as a successor to 9K35 Strela-10, and is a cheaper alternative to the Tor missile system and Pantsir-S1. Designed to work in passive mode with the help of different imaging systems (thermal camera, TV camera) and a laser rangefinder in order to find and engage a target, Sosna-R is able to operate effectively under the control of various types of old, modern and prospective battery command posts, the most preferable of which is the FPU "Assembly-M1-2" (9S80M1-2) and is resistant to jamming. It uses a tracked SAM launcher with 12 missiles of 10 km range and altitude of 5 km, based on the MT-LB vehicle. Those 12 missiles can be reloaded in 12 minutes.^{[6][7]} The serial production version is based on the BMP-3 IFV.^{[8][9]} The Sosna can also deliver fire while on the move.^[10] The system has a crew of two: the driver and gunner.

The system is armed with 12 9M340 missiles that are held in their transport-launch containers and arranged in two banks of six.^[11] Sosna-R beam rider 38 kg missiles are two-stage, capable of destroying aircraft, helicopters, cruise missiles, aerial bombs, small-sized air attack weapons including elements of high-precision weapons and light armored vehicles. The missile can change direction mid-flight.^{[12][13]} It also has a combined impact/proximity laser fuze. Its payload is made up of two warheads weighing a total of 7 kg. The fragmented-rod warhead is designed for proximity detonation when flying close to the target, while the armor-piercing/fragmentation (AP-Frag) warhead goes off on impact.^[14] The system has high resistance to jamming and besides its passive mode of operation it can be equipped with an additional small-size acquisition radar.

The Palma anti-aircraft artillery system features integrated armament that comprises eight Sosna-R precision surface-to-air missiles with laser beam guidance and two AO-18KDE rapid-fire guns. As a defense component, the Palma is designated to defend both ships and naval bases, coastal infrastructure and aerodromes, and also other vital facilities.^[15]

Range

Within 8–10 ($\leq$ 20) km.

Variants

- Sosna A weapon module for MBT, IFV, APC, AFV
- Sosna RA for AFV, MBT, IFV, APC
- Palash CIWS
- Palma CIWS - the deliveries of the systems to the Russian Navy and foreign navies started in 2011.^{[16][17]}
- Ptitselov CIWS is an air transportable version based on the BMD-4M chassis and it can also be dropped by parachute^[18]

Operators

Current operators



- Palma-SU CIWS used on Gepard 3.9 frigates.

Potential operator



- Potential Palma CIWS acquisition reported.^[19]

See also

- Hermes (missile)

References

1. "Sosna short-range air defense missile system" (https://www.armyrecognition.com/russia_russian_missile_system_vehicle_uk/sosna_short-range_air_defense_missile_system_technical_data_sheet_specifications_pictures_video_11312155.html).

9M337 Sosna-R



Sosna surface-to-air missile system.

Type	Surface to air missile
Place of origin	Russian Federation
Service history	
In service	May 2019
Used by	<u>Russian Federation</u>
Production history	
Designed	2017
Manufacturer	<u>High Precision Systems</u>
Produced	2022
Variants	See variants
Specifications	
Warhead	1× Fragmented-rod warhead 1× AP-Frag warhead
Detonation mechanism	Combined impact/proximity laser fuze
Operational range	10 km
Flight altitude	5 km
Maximum speed	Mach 2.6
Guidance system	<u>Laser guidance</u> <u>Radio guidance</u>
Launch platform	Sosna vehicle Ptitselov vehicle Palma CIWS Palash CIWS

2. "9K338 9M342 Igla-S / SA-24 Grinch" (<https://www.globalsecurity.org/military/world/russia/9m337.htm>).
3. "Новейший ЗРК "Багульник" успешно прошел испытания" (<https://www.vesti.ru/doc.html?id=3001367>). *vesti.ru*. Retrieved 21 January 2019.
4. ШАГвперед (8 April 2018). "Новейший российский ЗРК "Багульник" " (<https://www.youtube.com/watch?v=gQozwtvSAiQ>). Retrieved 21 January 2019 – via YouTube.
5. "Tłumacz Google" (https://translate.google.pl/translate?sl=auto&tl=en&js=y&prev=_t&hl=pl&ie=UTF-8&u=http%3A%2F%2Fwww.armstrade.org%2Finclude%2Fperiodics%2Fnews%2F2018%2F0822%2F095548302%2Fdetail.shtml&edit-text=&act=url).
6. Administrator. "Sosna short-range air defense missile system technical data sheet specifications pictures video 11312155 - Russia Russian missile system vehicle UK - Russia Russian army military equipment vehicles UK" (https://www.armyrecognition.com/russia_russian_missile_system_vehicle_uk/sosna_short-range_air_defense_missile_system_technical_data_sheet_specifications_pictures_video_11312155.html). *www.armyrecognition.com*.
7. Pike, John. "9K338 9M342 Igla-S / SA-24 Grinch" (<https://www.globalsecurity.org/military/world/russia/9k338.htm>).
8. "Army 2019: Nudelman displays Sosna missile air defense system and BMP-3 with add-on armor | Army-2019 News Russia Online Show Daily Media Partner | Defence security military exhibition 2019 daily news category" (https://www.armyrecognition.com/army-2019_news_russia_online_show_daily_media_partner/army_2019_nudelman_displays_sosna_missile_air_defense_system_and_bmp-3_with_add-on_armor.html).
9. "Janes | Latest defence and security news" (<https://www.janes.com/article/89554/army-2019-russia-unveils-new-sosna-variant-on-bmp-3-chassis>).
10. "Russian hi-tech firm showcases latest Sosna short-range air defense missile system" (<http://tass.com/defense/1018157>). TASS. Retrieved 21 January 2019.
11. "Army 2018: Russia unveils Sosna SHORAD system production version - Jane's 360" (<https://www.janes.com/article/82554/army-2018-russia-unveils-sosna-shorad-system-production-version>). *www.janes.com*. Retrieved 21 January 2019.
12. "SOSNA-R Air defense System" (<https://web.archive.org/web/20180106064245/http://defense-update.com/products/s/Sosna-R.htm>). *defense-update.com*. Archived from the original (<http://defense-update.com/products/s/Sosna-R.htm>) on 2018-01-06. Retrieved 2018-01-05.



The Palma-SU CIWS on Vietnamese Gepard 3.9 frigates using Sosna-R missiles.

13. "Sosna-R" (http://www.deagel.com/Defensive-Weapons/Sosna-R_a001903001.aspx). *www.deagel.com*. Retrieved 21 January 2019.
14. "Russian Sosna air defense missile system unveiled at Army-2018 | Army-2018 News Russia Online Show Daily | defense security exhibition 2018 pictures gallery" (https://armyrecognition.com/army-2018_news_russia_online_show_daily/russian_sosna_air_defense_missile_system_unveiled_at_army-2018.html). *Armyrecognition.com*. 2018-08-21. Retrieved 2019-01-21.
15. "Hi-tech firm delivers ten seaborne air defense systems to Russian Navy and foreign fleets" (<https://tass.com/defense/1068346>).
16. "Janes | Latest defence and security news" (<https://www.janes.com/article/89853/imds-2019-high-precision-systems-delivers-palma-aagms-to-russian-navy-foreign-customers>).
17. "ЦАМТО // Глава «Рособоронэкспорта» Александр Михеев: комплекс ПВО «Пальма» впервые интегрируют на иностранный корабль" (<https://armstrade.org/includes/periodics/news/2021/0825/105064196/detail.shtml>).
18. "Russia developed parachute-dropped Ptitselov SAM for Airborne units" (<https://bulgarianmilitary.com/2023/05/15/russia-developed-parachute-dropped-ptitselov-sam-for-airborne-units/>). 15 May 2023.
19. "Egypt Navy, Almaz-Antey, and Palma air defense systems" (<https://tacticalreport.com/news/article/57803-egypt-navy-almaz-antey-and-palma-air-defense-systems>). *Tactical Report*. 2021-09-28. Retrieved 2021-12-31.

Retrieved from "https://en.wikipedia.org/w/index.php?title=9M337_Sosna-R&oldid=1318208180"